Non-invasive multi-classifier for diagnosis of common skin cancers, via adhesive tape sampling

Table 1 Patient and lesion summary characteristics

	AK (N=73)	BCC (N=63)	Melanoma (N=75)	Nevus (N=83)	SCC (N=90)	Total (N=384)

Melanocytic lesions: Nevus (benign), Melanoma (cancer)
Keratinocytic lesions: Actinic keratosis (AK, benign pre-cancerous), squamous cell cancer (SCC), basal cell cancer (BCC)

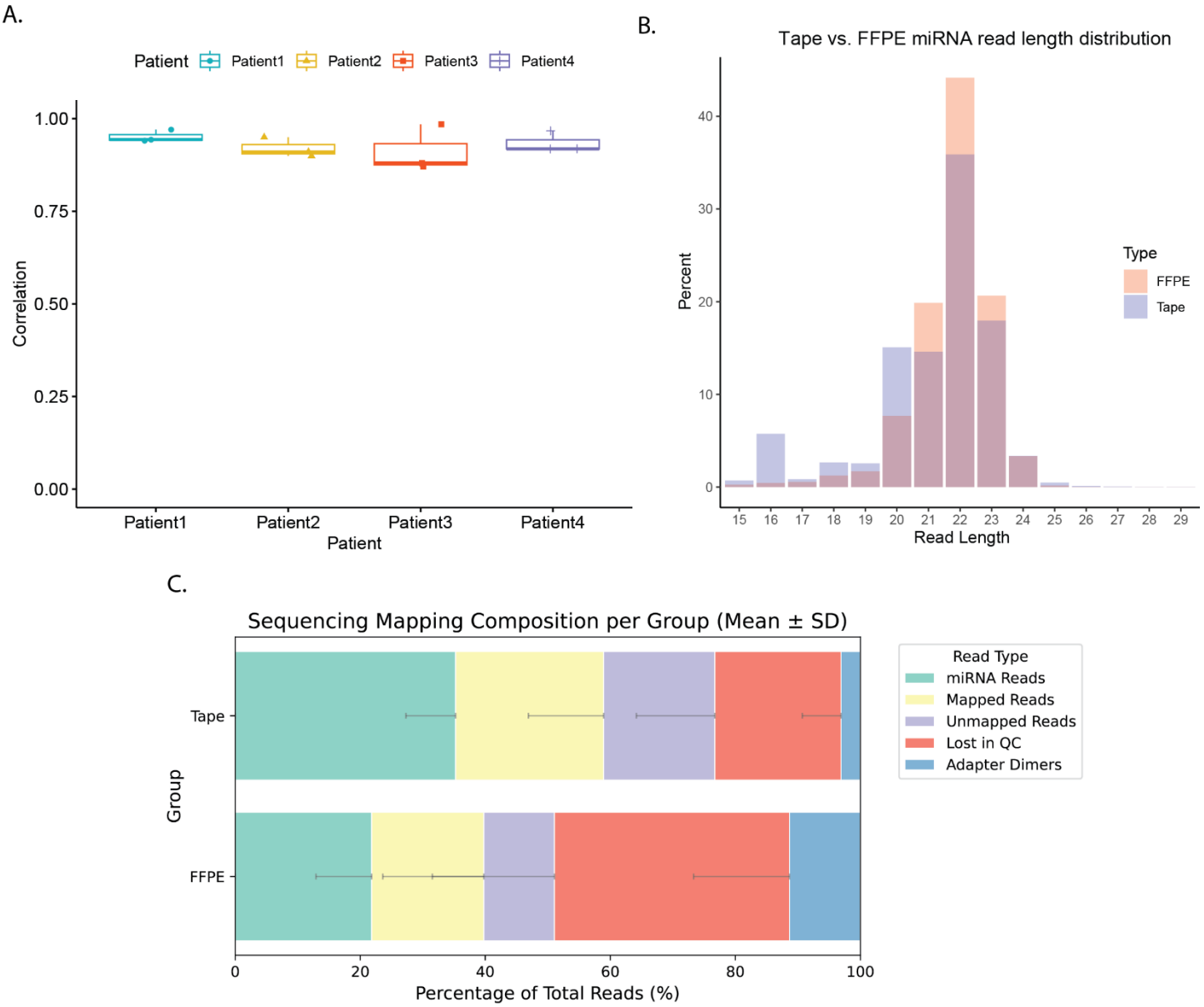

Figure 1. Tape collected miRNA reproducibility and stability

A) Plot showing average Spearman correlation for miRNA collected by serial tape sampling of single nevi on different individuals.

B) Read length distribution of representative miRNA isolated from formalin fixed paraffin embedded (FFPE) samples (pink) compared with miRNA collected by tape sampling (blue). Shown is the distribution of average percentage of the lengths of mapped miRNA reads plotted for a representative sample of 10 previously published FFPE[35] and 10 tape samples.

C) Sequencing statistics comparing representative FFPE samples to tape samples, showing tape samples had higher percentage of miRNA reads and lower percentage of adapter dimers.

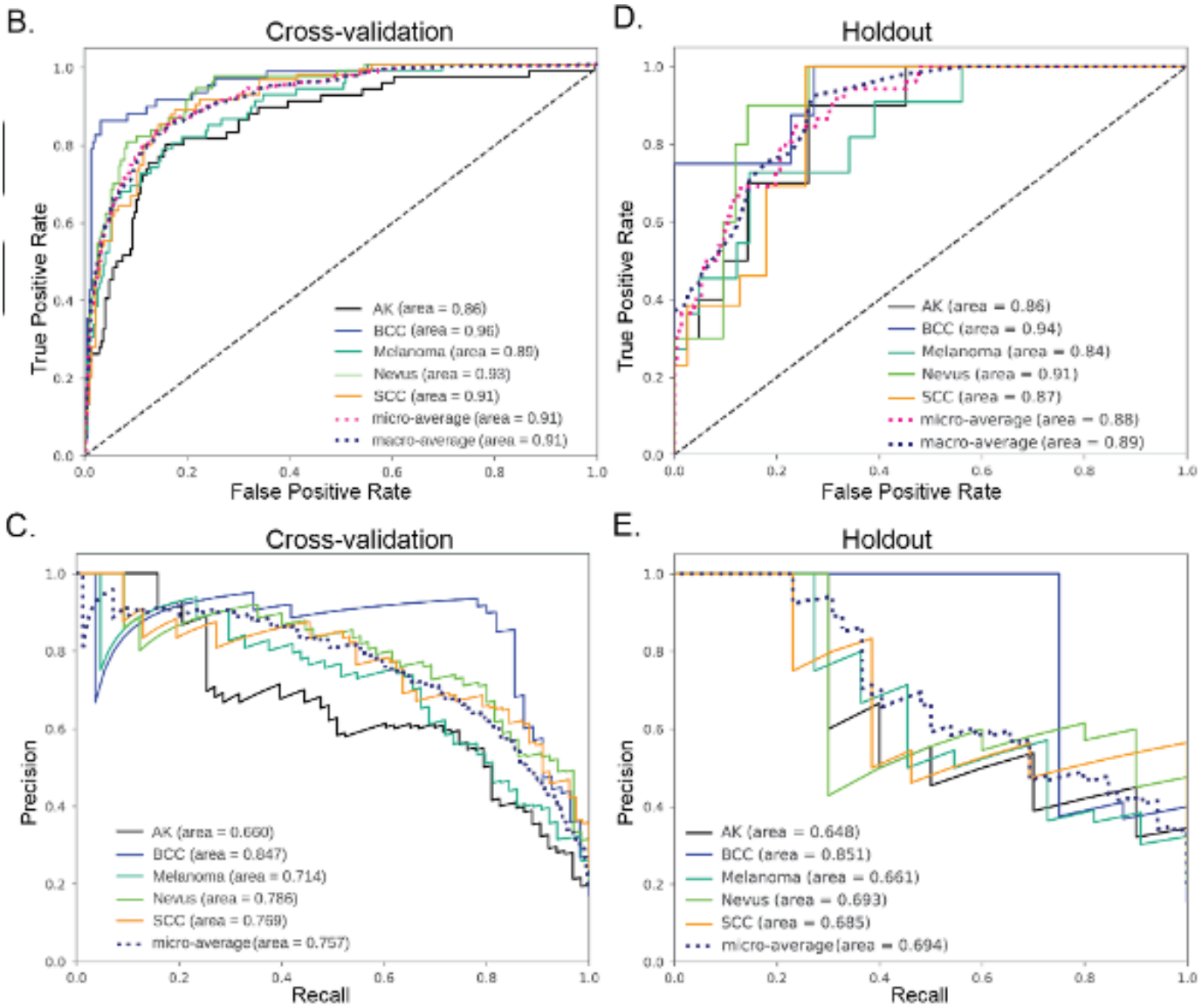

Figure 2. Model development and performance

B) AI neural network model AUROC cross-validation performance on training data. C) Precision recall curve cross-validation performance on training data. D) Model performance on hold-out data. E) Precision recall curve on hold-out data. AK, actinic keratosis; SCC, squamous cell cancer; BCC basal cell cancer

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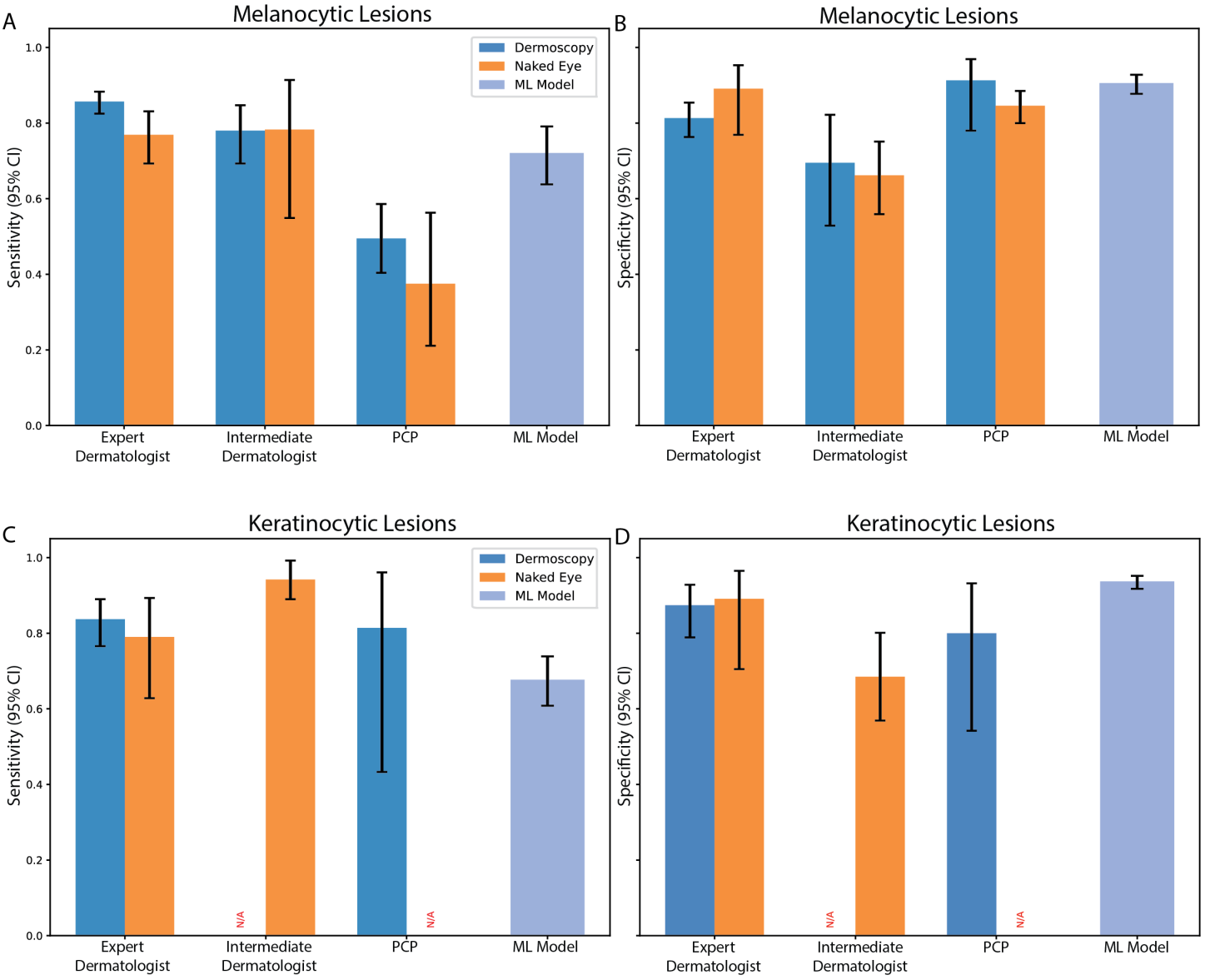
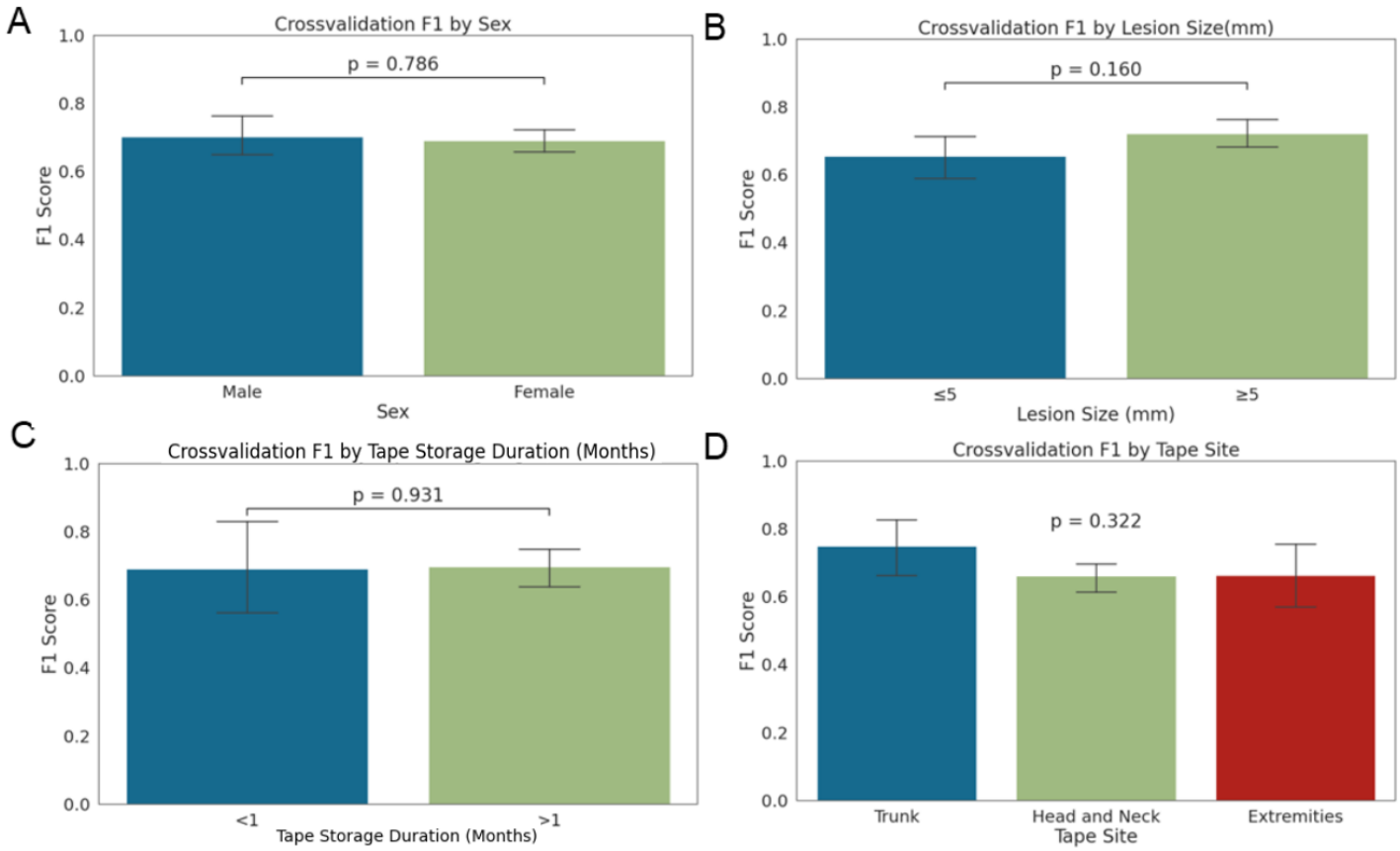
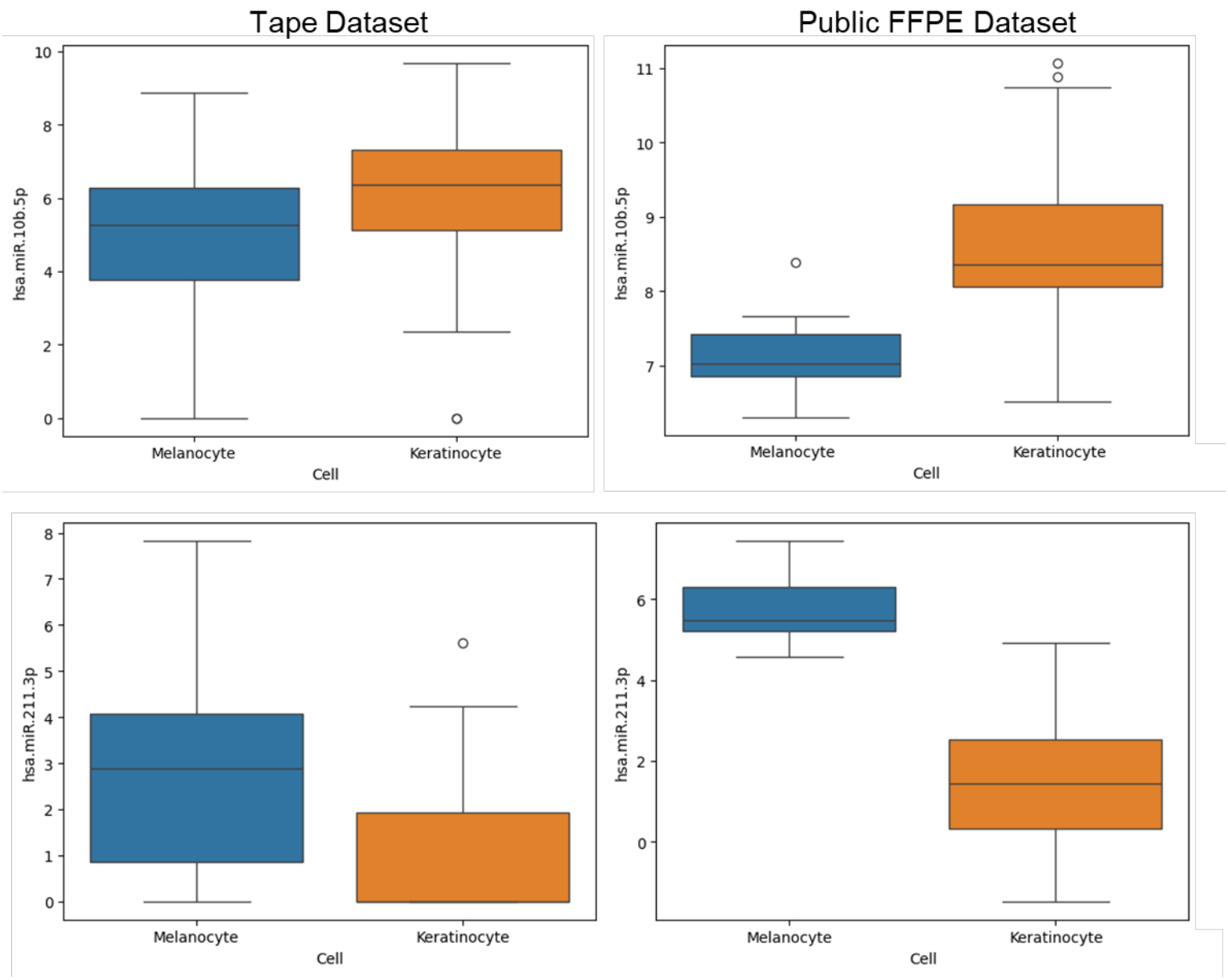


Figure 4. Model performance is comparable to PCPs' and dermatologists' performance
Sensitivity and specificity for the classifier calculated from aggregating cross validation performance on either melanocytic samples (nevus and melanoma; A and B) or keratinocytic samples (AK, SCC, and BCC; C and D) compared to experienced (≥ 2 years of practice), inexperienced dermatologists (<2 years of practice) and PCPs with (blue) or without (green) the use of a magnifying dermatoscope. AK, actinic keratosis; SCC, squamous cell cancer; BCC basal cell cancer; PCP, primary care physicians.

Supplement



Supplement Figure 2. Cross validation F1 performance stratified by sample characteristics
F1 score is compared between different subsets of samples to determine if a possible confounding variable affects performance. No significant difference observed by A) patient sex B) lesion size C) duration of sample storage and D) tape collection site. P value determined by T-test for two group or ANOVA for three group comparison.



Supplement Figure 3. miRNA is differentially expressed in tape samples in a similar manner to public datasets.
miRNA miR.10b.5p and miR.211.3p, (representative of other miRNAs), are differentially expressed in a similar manner in both our tape samples and in reanalyzed public datasets derived from formalin fixed paraffin embedded samples,[49–52] when comparing melanocytic and keratinocytic lesions.